



AI CAREERS FOR WOMEN LINKEDIN FELLOWSHIP

PERSONAL FINANCIAL PLANNING_ FORECASTING MODEL

Project Domain

[Excel / Power BI / Power BI with Copilot / Claude

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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3. Problem Statement

Personal finance management is increasingly complex due to rising living costs, unpredictable income growth, and high debt burdens. The dataset reveals that most individuals, regardless of age or income level, fall into the **“Risky” financial health category**, with low savings ratios and high expense or loan ratios. Data-related challenges include incomplete emergency funds, inconsistent savings behaviour

and difficulty in integrating external economic factors (inflation, interest rates, market volatility) with personal records. These challenges highlight the need for a forecasting model that can process diverse financial indicators and predict long-term outcomes.

Gap / Inefficiency

Current personal finance tools are descriptive, focusing on past spending or current balances. They rarely provide **forward-looking insights** or integrate multiple dimensions such as savings, debt, investments, and emergency funds. The dataset shows that while future savings projections exist, they are static and do not account for behavioural changes or external shocks. This project aims to fill that gap by building a **predictive, scenario-based model** that forecasts financial health scores over time and suggests actionable improvements.

Stakeholders & Value

- **Individuals & Families:** Gain clarity on their financial trajectory, with personalized forecasts and recommendations.
- **Financial Advisors:** Use the model to provide data-driven, holistic advice.

Forward-Looking Insights

Forward-looking insights are analytical outputs that show *what is likely to happen* in the future, not just *what has already happened*.

Current personal finance tools are **descriptive**, focusing only on past spending or current balances. They fail to provide **forward-looking insights** that anticipate future risks and opportunities. For example:

- They don't forecast how today's debt obligations will impact savings five years from now.
- They don't simulate external shocks such as inflation, interest rate hikes, or market volatility.

- They don't account for behavioural changes like increasing savings or restructuring loans.

Without these predictive capabilities, individuals remain reactive, often realizing financial stress only after it occurs. This gap highlights the need for a **scenario-based forecasting model** that can project financial health scores over time, simulate "what-if" situations, and provide actionable recommendations before risks materialize.

4. Objectives

Data Acquisition & Exploration

- Collect and consolidate raw personal financial datasets (income, expenses, savings, loans, investments, emergency funds).
- Perform deep exploratory analysis to uncover structural patterns, anomalies, and data quality issues that could distort forecasting accuracy.

2. Data Cleaning & Standardization

- Implement rigorous preprocessing: remove duplicates, handle missing values, correct inconsistent formats, and normalize financial ratios.
- Establish a standardized data pipeline to ensure reliability and scalability for future datasets.

3. Domain-Driven Financial Analysis

- Apply advanced Excel formulas, pivot tables, and Power BI measures to derive key ratios (Savings Ratio, Expense Ratio, Loan Ratio).
- Use AI-driven prompt engineering to generate scenario-based insights, simulating market shocks, lifestyle changes, and debt restructuring.

4. Predictive Modelling & Forecasting

- Develop a robust forecasting engine that projects savings, debt, and financial health scores over 5+ years.
- Integrate Monte Carlo simulations and regression models to capture uncertainty and provide realistic outcomes.

5. Interactive Dashboard & Reporting

- Build a dynamic, user-friendly dashboard that visualizes financial trajectories, risk categories, and improvement strategies.
- Deliver structured reports with actionable insights tailored for individuals, advisors, and fintech stakeholders.

6. AI Integration for Acceleration

- Leverage Microsoft Copilot and AI chatbots to automate repetitive analysis, generate financial narratives, and enhance stakeholder communication.
- Deploy AI-assisted recommendations that adapt to user behaviour and external economic indicators.

7. End-to-End Documentation & Deployment

- Document the complete project lifecycle: data collection, preprocessing, modelling, visualization, and AI integration.
- Upload all files, scripts, and dashboards to GitHub for transparency and collaboration.
- Record a professional demonstration video showcasing the model's capabilities and business value.

8. Project-Specific Objective (Unique to Domain & Tool)

- **Design a Financial Health Index (FHI)** that integrates multiple ratios (savings, expenses, loans, investments, emergency funds) into a single predictive score.
- Use Excel/Power BI as the primary tool to calculate, visualize, and track this index, while AI tools generate personalized improvement strategies.
- Position the FHI as a **benchmark metric** that can be adopted by fintech platforms and advisors to evaluate and forecast personal financial resilience.

5. Methodology / Implementation Plan

Provide a simple overview of how the project will be carried out. The following five-phase workflow applies to all projects under this fellowship program.

#	Phase	Description
1	Data Collection & Exploration	1 Obtain the raw dataset (Excel file with fields such as income, expenses, savings, loans, investments, emergency funds, ratios, and financial health scores). • Explore the dataset to understand its structure, field definitions, and row count. • Identify data quality issues such as inconsistent ratios, missing emergency fund values, or unrealistic savings projections. • Define the use case: forecasting personal financial health over 5+ years for individuals and families, with the target audience being fintech platforms, advisors, and end-users seeking proactive financial planning.
2	Data Cleaning & Preprocessing	Remove duplicate records and handle missing or null values in savings, expenses, and loan fields. • Correct data types (e.g., numeric vs. text inconsistencies in financial ratios). • Standardize formats for monetary values, percentages, and categorical fields (e.g., “Risky” score category). • Use Excel functions (IF, IFERROR, Text-to-Columns) and Power Query to ensure clean, reliable inputs for forecasting.

3	Analysis & AI Integration	Apply pivot tables, Excel formulas (SUMIF, COUNTIF, VLOOKUP) and Power BI DAX measures to derive insights such as: ○ Savings Ratio, Expense Ratio, Loan Ratio trends. ○ Financial Health Score distribution across age groups and income levels. ○ Future savings projections under different scenarios. • Integrate AI tools (Microsoft Copilot / AI Chatbot) to generate narratives, summaries, and personalized recommendations from the dataset. • Use AI prompt techniques to simulate scenarios (e.g., job loss, debt restructuring, increased savings rate) and produce actionable insights.
4	Visualization / Output	Build an interactive dashboard in Power BI or Excel with: ○ KPI cards for Savings Ratio, Expense Ratio, Loan Ratio, and Financial Health Score. ○ Charts showing savings growth, debt reduction, and net worth projections over 5 years. ○ Slicers for filtering by age, income, or risk category. • For AI-assisted outputs, compile generated content such as financial health reports, improvement strategies, and personalized recommendations. • Document all AI prompts used alongside their outputs for transparency.
5	Documentation & Submission	Write a comprehensive project report covering data collection, preprocessing, analysis, visualization, and AI integration. • Upload all files (Excel dataset, Power BI dashboard, scripts, and documentation) to GitHub for collaboration and evaluation.

	- Record a professional demonstration video showcasing the dashboard, forecasting model, and AI-generated insights. - Submit GitHub link, video link, and dashboard link for final review.
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6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC: HP Victus with Intel Core i5, minimum 16 GB RAM (8 GB recommended)
- Operating System: Windows 11 Home (fully compatible with Excel, Power BI, and Copilot).
- Microsoft 365 or Excel 2019+
- Power BI Desktop — latest version (free download from Microsoft) — where applicable
- Microsoft Copilot — Accessible via Microsoft 365 subscription, runs smoothly on Windows 11.
- AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — Compatible with Copilot Web, ChatGPT, and Google Gemini for prompt-based analysis
- Internet connection — ☑ Required for Copilot, AI chatbot integration, and GitHub uploads.
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. DATA SET

The dataset was generated using **synthetic but realistic financial data**.

- Each row represents a customer profile with demographic and financial attributes.
- Ratios (Savings Ratio, Expense Ratio, Loan Ratio) and Financial Health Scores were calculated using domain-relevant formulas.
- Future savings projections (baseline and improved scenarios over 5 years) were simulated to reflect realistic financial planning outcomes.

- **Structure:**

- **Rows:** Each row corresponds to one individual's financial profile.
- **Columns:** Include demographic fields (Age), financial fields (Income, Expenses, Savings, Loans, Investments, Emergency Fund), behavioural fields (Credit Card Spending), and calculated fields (Ratios, Financial Health Score, Score Category, Future Savings).
- **Categories:** Most individuals fall under the “Risky” category, highlighting the dataset’s focus on financial instability and the need for forecasting.

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	Week 1
2	Data cleaning, preprocessing, and validation / Prompt design	Week 2
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	Week 2 – 3
4	Dashboard / output build, testing, and review	Week 3
5	Report writing, GitHub upload, and video recording and submission	Week 4

7. Expected Outcomes / Conclusion

At the heart of the model is the **Financial Health Score (FHS)**, a composite index ranging from 0 to 100. It is calculated using five normalized sub-scores: savings rate, debt ratio, emergency fund adequacy, income stability, and investment adequacy. Each factor carries a weight, with savings and debt contributing the most, because they are the strongest predictors of long-term resilience.

Once computed, the FHS places individuals into one of four categories: Healthy, Moderate, Vulnerable, or Risky. This classification is not just descriptive—it triggers specific actions, such as improvement cards, advisor alerts, or intervention plans, ensuring that insights lead directly to solutions.

The dataset powering this model includes both **personal records**—income, expenses, savings, loans, investments—and **external signals** like inflation, interest rates, and market volatility. These inputs are cleaned, normalized, and engineered into features such as savings gaps and debt-to-asset ratios.

The processing layer runs three parallel modules. First, the **FHS Scoring Engine** calculates the current financial health. Second, the **Forecasting Engine** uses advanced machine learning—XGBOOST for short-term projections and LSTM neural networks for long-term trajectories. Third, the **Scenario Simulation Engine** runs Monte Carlo simulations to stress-test thousands of possible futures, from job loss to salary hikes, from interest rate changes to debt repayment strategies.

All outputs feed into a **dynamic dashboard** that presents scores, forecasts, risk alerts, and personalized action cards. These insights are then delivered to stakeholders: individuals and families receive tailored improvement plans, advisors gain multi-client risk profiles, and fintech platforms can integrate predictive analytics through APIs.

8. REFERENCES

Core Fellowship & Institutional References

EDUNET Foundation. AICW Fellowship Program Guidelines: AI Careers for Women Project Framework. 2026. Available at: <https://www.edunetfoundation.org>

LinkedIn Learning. AI and Data Analytics for Business Professionals. 2026. Available at: <https://www.linkedin.com/learning>

Technical Software & AI Documentation

- **Microsoft Corporation.** Microsoft Excel: Advanced Formulas and Data Analysis Tool pak Documentation. Available at: <https://support.microsoft.com/en-us/excel>
- **Microsoft Corporation.** Power BI: DAX (Data Analysis Expressions) for Financial Modelling. Available at: <https://learn.microsoft.com/en-us/power-bi/>
- **Microsoft Corporation.** Microsoft Copilot: Leveraging Generative AI for Automated Financial Insights. Available at: <https://copilot.microsoft.com>

